Finding min and max concurrently

Suppose you have an array of n distinct numbers. In the ICS 30-series, you learned how to find the min or max of such an array. Suppose you wanted to find *both* – the min and max.

One way to do this would be to find the min; this takes $n-1$ comparisons. You could then output and delete the min element and find the max of what remains, taking $n-2$ comparisons, for a total of $2n-3$ comparisons.

Question 1. Can you find a way to find both using *strictly fewer than* $2n-3$ *comparisons*? Note that we are measuring the actual number of comparisons, not the growth rate of your function.

If you are having trouble starting, you may assume n to be odd or even (your choice).

Question 2. Could anyone solve this problem in fewer comparisons than your solution uses?

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